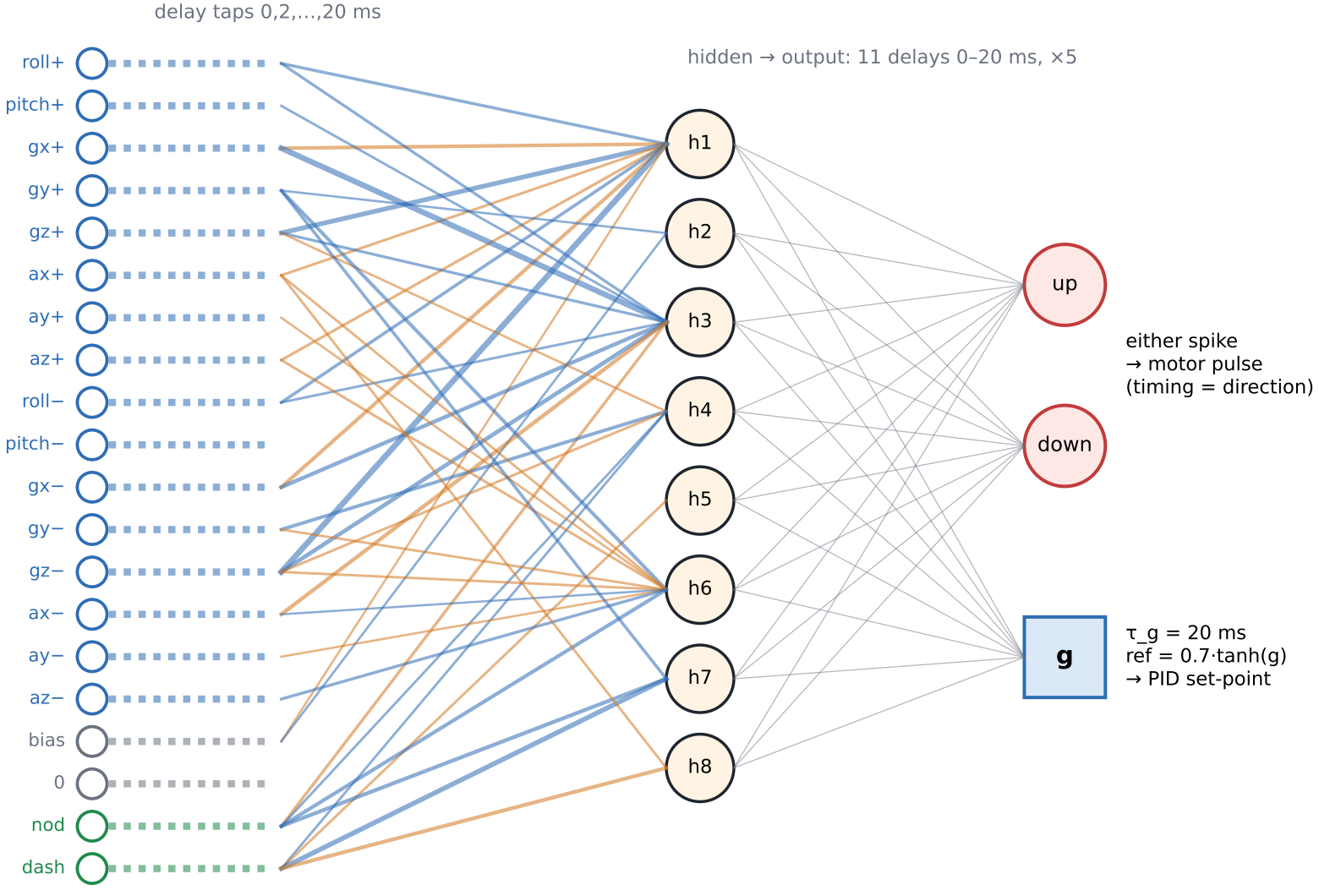
Descending SNN: 20 inputs × 11 delays → 8 LIF → 2 spiking outputs + graded unit (2035 parameters)



Edges drawn: input→hidden weights summed over delays, |w| ≥ 25% of max (orange +, blue −), round-5 seed 4.

LIF dynamics (2 × 1 ms substeps per tick)

$$I_h = \sum_i \sum_k W^{in}_{h,i,k}\, x_i(t - 2k\,\text{ms}) + b_h$$
$$c_h \leftarrow a_s c_h + (1 - a_s) I_h$$
$$v_h \leftarrow a_m v_h + (1 - a_m) c_h$$
$$s_h = [v_h \geq 1], \quad v_h \leftarrow v_h - s_h$$
$$I_o = 5 \sum_h \sum_d W^{out}_{o,h,d}\, s_h(t - d) + b_o$$
$$ref = 0.7 \tanh(g)$$

parameters (searched by ES)

W_in	8×20×11	1760
b_h	8	8
W_out	2×8×11	176
b_o	2	2
W_g	8×11	88
b_g	1	1
total		**2035**

τ_m = 1.26 ms, τ_s = 1.04 ms
(learned in imitation, fixed in ES)